FRIDAY 23 OCTOBER 2026 · 55 MINUTES TOTAL

Pure Mathematics + Linear Algebra

Week 5 combined problem tutorial · Group 07

> 周四播放视频后未讲题，本次把两门课合在一个时段。以短思考、重点板书、相邻小题口头核对为主；不照抄两套完整课时。下列 24+23 分钟是讲题预算，包含学生思考时间。

Elapsed	What to do	Hard stop
00–05	Key ideas (2 min), arrow and vector visuals (3 min). Ask one diagnostic question.	Start Pure at 05:00
05–29	Pure: classify functions → a proof → images/preimages → intersections → composition.	Switch at 29:00
29–52	Linear: dependence → find bases in three forms → complex examples → dimension and basis logic.	Exit check at 52:00
52–55	Two short exit questions and consolidation-week goal.	Finish at 55:00

Opening words: “Yesterday we used a video to think about mathematics more broadly. Today we will work through the problem-solving tools from both courses. For each topic, try a first step, then compare your reasoning with the board.”

> 每题先留 20–40 秒写第一步；一名学生回答后直接板书。到点必须切换。赶时间时压缩重复计算和第二个例子，不删证明中的量词、定义域或“生成且独立”这两个条件。

If the room is behind: at minute 22, finish the current Pure proof and give the remaining Pure conclusions orally; at minute 45, stop expanding new bases and move to the dimension table. Reserve questions are for lookup, not additional promised teaching time.

Scope: Pure PS5 and the Week 5 Linear Algebra tutor guide. Inverse-function material was scheduled for Friday 13:00 and is marked as preview/reserve. Linear lecture progress was confirmed through §6.30 on 20 October; the end of Chapter 6 was planned for Thursday, so ask a quick opening check. Lecturer hand-in model solutions are not included.

START HERE · 00–05 MINUTES

Key ideas before the questions

Pure Mathematics + Linear Algebra · Week 5 · Keep this page beside your working.

1 · Functions: always state the domain and codomain

For $f : X \to Y$: **injective** means $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$; **surjective** means every $y \in Y$ has some $x \in X$ with $f(x) = y$. **Bijective** means both.

For $A \subseteq X$, $B \subseteq Y$:

$$f[A] = \{f(x) : x \in A\}, \qquad f^{-1}[B] = \{x \in X : f(x) \in B\}.$$

An image follows inputs forward; a preimage collects *all* inputs landing in the target set. Preimages exist even when no inverse function exists.

For $f : X \to Y$, $g : Y \to Z$, $(g \circ f)(x) = g(f(x))$: apply f first. The composition of two surjections is surjective.

Proof checklist: injectivity → start with equal outputs; surjectivity → choose an arbitrary target and construct a legal input; set equality → prove both inclusions; a false universal claim → give one valid counterexample.

Watch: $f[A \cap B] \subseteq f[A] \cap f[B]$ always; equality holds if f is injective. Changing the codomain can change surjectivity.

2 · Vector spaces: always state the scalar field

For $v_1, \ldots, v_k \in V$ over a field K:

$$\operatorname{span}_K\{v_1, \ldots, v_k\} = \left\{ \sum_{j=1}^{k} a_j v_j : a_j \in K \right\}.$$

Independent: $\sum a_j v_j = 0$ forces every $a_j = 0$. **Dependent:** some such relation has at least one nonzero coefficient. A **basis** is both independent and spanning; $\dim_K V$ is the number of vectors in any basis.

If $\dim V = n$: fewer than n vectors cannot span; more than n are dependent; exactly n form a basis if either independence or spanning is proved. Size alone is not enough.

Find a basis: parameters → separate coefficients and check independence; equations → solve with free parameters; a spanning set → remove redundant vectors, or take pivot columns from the *original* matrix. To extend an independent set, add vectors outside its current span.

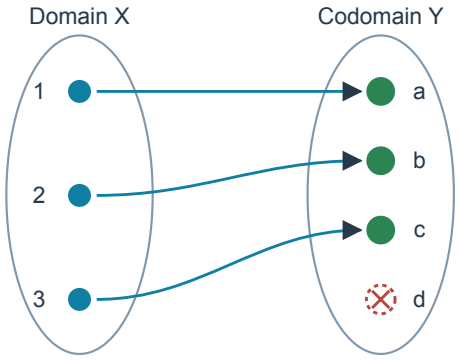
Watch: ambient coordinate count is not subspace dimension. Real and complex scalars give different independence tests. The zero space has empty basis, not $\{0\}$.

VISUAL REVIEW 1 · FUNCTIONS · USE WITHIN THE OPENING 5 MINUTES

Follow the arrows

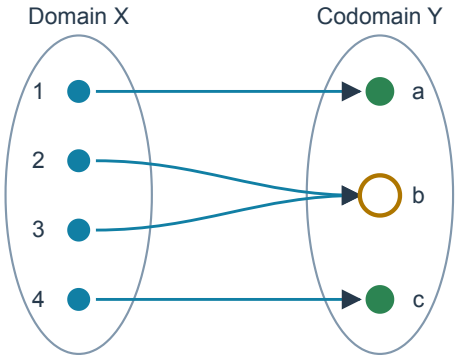
Every input must have **exactly one** arrow. Then ask: do inputs collide, and is any target missed?

Injective only



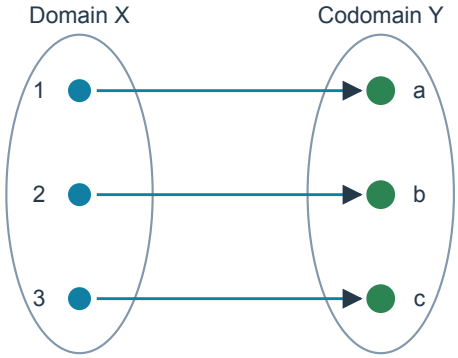
No collision, but one target is missed.
Injective: yes · Surjective: no

Surjective only



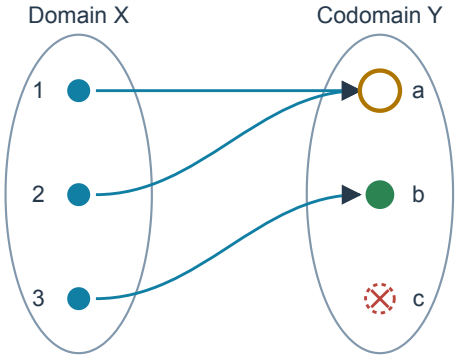
Every target is hit, but two inputs collide.
Injective: no · Surjective: yes

Bijective



Every target has exactly one preimage.
Injective: yes · Surjective: yes

Neither



A collision and a missed target.
Injective: no · Surjective: no

Blue arrows = function values · Yellow ring = collision · Dashed ring with × = missed target.

Injective: each target has at most one preimage.
Surjective: each target has at least one preimage.
Bijective: each target has exactly one preimage.

Ask the room: “If we remove the unused target from the first codomain, what changes?”

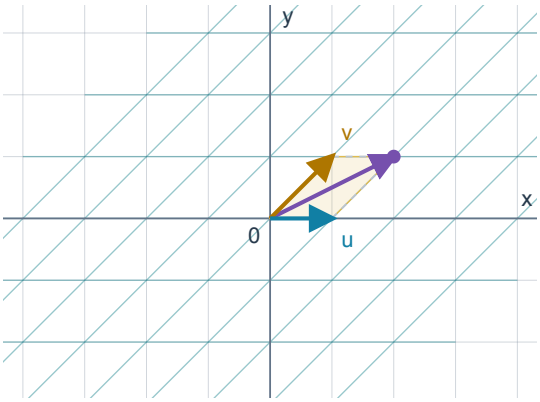
These are separate finite examples. All displayed targets belong to the stated codomain; changing the codomain changes the question.

VISUAL REVIEW 2 · LINEAR ALGEBRA · ALL SCALARS ARE REAL

Two vectors: plane or line?

Fix $u = (1, 0)$, let $v = (1, t)$, and form $au + bv = (a + b, bt)$.

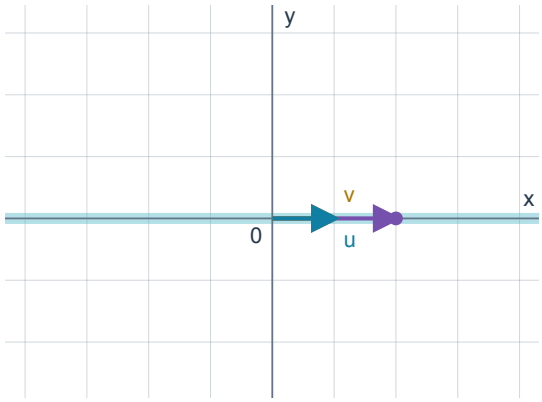
Independent: t = 1



Span: the whole plane

The two directions generate every point of $\mathbb{R}^2$. Each point has unique coefficients.

Collinear: t = 0



Span: one line through 0

Now $u = v$. Only the horizontal axis is reached, and coefficients are not unique.

Blue = u · Yellow = v · Purple = a u + b v · Dashed lines = vector addition. The highlighted parallelogram collapses when t = 0. The grid is only a visible sample; the span is unbounded.

The bridge between both courses

Consider the coefficient map $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(a, b) = au + bv$.

Independent vectors ⇔ **injective** T: unique coefficients.
Spanning $\mathbb{R}^2$ ⇔ **surjective** T: every target is reached.
A basis of $\mathbb{R}^2$ ⇔ **bijective** T: both properties.

Vector space check. A subspace of $\mathbb{R}^2$ contains 0 and is closed under addition and scalar multiplication. A line through 0 is a subspace; the line $y = 1$ is not, since it omits 0.

At t = 0: the list (u, v) is dependent. The singleton $\{u\}$ is a basis of its one-dimensional span, but it is not a basis of $\mathbb{R}^2$.

Ask the room: “At t = 0, compare T(1,0) and T(0,1). Can any input reach (0,1)?”

05–29 · Pure Mathematics

Functions, image/preimage and proof.

MAIN ROUTE · 05–08 elapsed · 3 min

P1 · Quick diagnosis: injective versus surjective

Pure PS5 A9(a–c), A10(a–c)

A9. Let $f : \mathbb{R} \to \mathbb{R}^2$, $f(x) = (2x, x + 3)$. Find $f(0)$, $f(1)$; prove that f is injective; show that $(4, 4)$ is not attained.

A10. Let $h : \mathbb{R}^2 \to \mathbb{R}$, $h(x, y) = x + y$. Find $h(0, 1), h(2, 3)$; use $(0, 0), (-1, 1)$ to test injectivity; prove surjectivity.

> **Say:** An injection never merges two inputs. A surjection reaches every point of the stated codomain. To disprove either one, a single counterexample is enough.

> 先让学生口答，再写两个定义的量词。不要用“输入维数小”替代证明。A10 用 h 是为避免与 A9 的 f 混淆。

A9. $f(0) = (0, 3)$, $f(1) = (2, 4)$. If $f(x_1) = f(x_2)$, equality of first coordinates gives $2x_1 = 2x_2$, hence $x_1 = x_2$: injective. If $f(x) = (4, 4)$, the first coordinate forces $x = 2$, but then the second coordinate is 5, a contradiction. Thus f is not surjective.

A10. $h(0, 1) = 1$, $h(2, 3) = 5$. The distinct inputs $(0, 0)$ and $(-1, 1)$ both map to 0, so h is not injective. For an arbitrary $u \in \mathbb{R}$, the input $(u, 0)$ satisfies $h(u, 0) = u$. Therefore h is surjective. Neither map is bijective.

MAIN ROUTE · 08–12 elapsed · 4 min

P2 · A proof using coordinates

Pure PS5 B3(a–c)

Let $F : \mathbb{R}^2 \to \mathbb{R}^3$ be $F(x, y) = (x + y, x - y, x^2 + y^2)$. Prove that F is injective and not surjective. Is it bijective?

> **Say:** We only need the first two coordinates to recover the input. For non-surjectivity, do not solve for every possible target: choose one impossible target.

> 连接线代：前两坐标形成可逆的 2×2 线性系统；第三坐标非线性，不要称整个映射为线性映射。全像公式只作追问。

Suppose $F(x_1, y_1) = F(x_2, y_2)$. The first two coordinates give

$$x_1 + y_1 = x_2 + y_2, \qquad x_1 - y_1 = x_2 - y_2.$$

Adding and subtracting gives $2x_1 = 2x_2$ and $2y_1 = 2y_2$. Hence $(x_1, y_1) = (x_2, y_2)$, so F is injective. The target $(0, 0, -1) \in \mathbb{R}^3$ is not attained, because $x^2 + y^2 \geq 0$ for all real inputs. Thus F is not surjective and therefore not bijective.

If asked for the whole image: writing the output as (u, v, w), the first two coordinates force $x = (u + v)/2$, $y = (u - v)/2$, so $w = (u^2 + v^2)/2$. Every point satisfying this equation is attained.

MAIN ROUTE · 12–17 elapsed · 5 min

P3 · Images and preimages: horizontal projection

Pure PS5 A2(a–b)

Let $f : \mathbb{Z}^2 \to \mathbb{Z}^2$, $f(m, n) = (m, 0)$, and

$$A = \mathbb{Z}_{>0} \times \{0\}, \quad B = \{0\} \times \mathbb{Z}, \quad C = \{(n, n) : n \in \mathbb{Z}\}.$$

Find $f[A], f[B], f[C]$, and $f^{-1}[A], f^{-1}[B], f^{-1}[C]$.

> **Say:** For an image, start with an allowed input and see where it lands. For a preimage, start with the target set and ask which inputs land inside it. Preimage notation does not assume that an inverse function exists.

> 画横轴、纵轴、对角线三条点集。特别强调 Z+ 的限制，以及原像≠逆函数。

The map keeps the first coordinate and replaces the second by zero. Thus

$$f[A] = \mathbb{Z}_{>0} \times \{0\} = A, \quad f[B] = \{(0, 0)\}, \quad f[C] = \mathbb{Z} \times \{0\}.$$

For preimages use $(m, n) \in f^{-1}[S] \iff (m, 0) \in S$. Membership of $(m, 0)$ in A means $m > 0$; membership in B means $m = 0$; membership in C means $m = 0$. The second input coordinate is unrestricted. Hence

$$f^{-1}[A] = \mathbb{Z}_{>0} \times \mathbb{Z}, \qquad f^{-1}[B] = f^{-1}[C] = \{0\} \times \mathbb{Z}.$$

Source correction: the supplied model solution mistakenly writes $f[A] = \mathbb{Z} \times \{0\}$. The question defines A using positive integers, so the correct image is A, not the whole horizontal axis.

MAIN ROUTE · 17–23 elapsed · 6 min

P4 · Why images do not always preserve intersections

Pure PS5 A5(a–c), B6, B7

A5. For $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, let $A = (-1, 0)$, $B = (0, 1)$. Find $f[A \cap B]$, $f[A]$, $f[B]$, and prove strict containment of the first set in $f[A] \cap f[B]$.

B6. If $f : X \to Y$ is not injective, construct $A, B \subseteq X$ with $f[A \cap B] \subsetneq f[A] \cap f[B]$.

B7. If f is injective, prove equality for all $A, B \subseteq X$.

> **Say:** An output can be reached from A and from B using two different inputs. Injectivity is exactly what forces those two witnesses to be the same input.

> 这里是 Pure 主证明。必须把 a∈A 与 b∈B 分开写，不能一开始就假设同一个 x。先例子、再抽象反例、最后双包含。

A5. $A \cap B = \varnothing$, so $f[A \cap B] = \varnothing$. Squaring maps both intervals onto $(0, 1)$: each $y \in (0, 1)$ has preimages $-\sqrt{y} \in A$, $\sqrt{y} \in B$, and all squares from those intervals lie in $(0, 1)$. Hence $f[A] = f[B] = (0, 1)$, and $\varnothing \subsetneq (0, 1)$.

B6. Since f is not injective, choose $a \neq b$ with $f(a) = f(b)$. Set $A = \{a\}$, $B = \{b\}$. Then $f[A \cap B] = \varnothing$, whereas $f[A] \cap f[B] = \{f(a)\}$, a nonempty set.

B7, first inclusion (no injectivity needed). If $y \in f[A \cap B]$, there is $x \in A \cap B$ with $f(x) = y$. This one input belongs to both sets, so $y \in f[A] \cap f[B]$.

Reverse inclusion (where injectivity is used). If $y \in f[A] \cap f[B]$, there are $a \in A$, $b \in B$ with $f(a) = y = f(b)$. Injectivity gives $a = b$. Thus $a \in A \cap B$ and $y = f(a) \in f[A \cap B]$. Both inclusions prove equality.

MAIN ROUTE · 23–26 elapsed · 3 min

P5 · Surjective maps compose

Pure PS5 A11

Suppose $f : X \to Y$ and $g : Y \to Z$ are surjective. Prove that $g \circ f : X \to Z$ is surjective.

> **Say:** Work backwards from an arbitrary target, but write the composition in the correct forward order. The intermediate point is chosen to hit your target; it is not arbitrary.

> 画 X→Y→Z；选点方向反着走，函数运算方向不变。

Take an arbitrary $z \in Z$. Since g is surjective, choose $y \in Y$ with $g(y) = z$. Since f is surjective, choose $x \in X$ with $f(x) = y$. Then

$$(g \circ f)(x) = g(f(x)) = g(y) = z.$$

Thus every target z is attained, which proves surjectivity.

MAIN ROUTE · 26–29 elapsed · 3 min

P6 · Four concrete image–preimage examples

Pure PS5 A6(a–d)

For $f : \mathbb{Z} \to \mathbb{Z}$, $f(x) = x^2$, find sets satisfying:

(a) $U_1 \subsetneq f^{-1}[f[U_1]]$; (b) $f[f^{-1}[V_1]] \subsetneq V_1$; (c) $U_2 = f^{-1}[f[U_2]]$; (d) $f[f^{-1}[V_2]] = V_2$.

> **Say:** Going out and back can add inputs that share the same output. Going back and out can lose target values that the function never reaches.

> 收尾检查四组，停在具体例子；B8(b) 的一般证明留学生自己完成。

(a) Take $U_1 = \{1\}$. Then $f[U_1] = \{1\}$, while $f^{-1}[\{1\}] = \{-1, 1\}$, so containment is strict.

(b) Take $V_1 = \{-1\}$. No integer square is negative, so $f^{-1}[V_1] = \varnothing$ and $f[\varnothing] = \varnothing \subsetneq \{-1\}$.

(c) Take $U_2 = \{-1, 1\}$. Its image is $\{1\}$, whose preimage is precisely U_2.

(d) Take $V_2 = \{1\}$. Its preimage is $\{-1, 1\}$, whose image is $\{1\} = V_2$.

Source correction: A6(b) in the supplied solution ends with $\{1\}$; it should end with $\{-1\}$. These are worked examples only; do not turn them into the complete proof of hand-in B8.

29–52 · Linear Algebra

Independence, spanning, basis and dimension.

MAIN ROUTE · 29–31 elapsed · 2 min

LA1 · Independence, spanning and a basis

Linear Algebra tutorial guide, Week 5, pp. 12–13; examples supplied for the guide's discussion prompts.

What do linear independence, spanning and a basis mean? Give an example in each of the four boxes below, with ambient space $\mathbb{R}^2$: independent/spanning, independent/not spanning, dependent/spanning, dependent/not spanning. Why are both conditions needed for a basis?

> **Say:** A basis must do two jobs: reach every vector and do so without redundancy. Keep the ambient space in view: one nonzero vector is a basis of its line, but not of the whole plane.

> 先用四格快速口答，别把"独立"和"生成"当成互为否定。强调都是相对于给定空间和数域。

A finite family $v_1, \ldots, v_k$ is **linearly independent** when $\sum_{j=1}^{k} \lambda_j v_j = 0$ implies every $\lambda_j = 0$; it is dependent when a nonzero coefficient tuple gives zero. It **spans** V when every vector of V is a linear combination of the family. A **basis** is an independent spanning set.

In $\mathbb{R}^2$	Spanning	Not spanning
Independent	$\{e_1, e_2\}$	$\{e_1\}$
Dependent	$\{e_1, e_2, e_1 + e_2\}$	$\{e_1, 2e_1\}$

Here $e_1 = (1, 0)^T$, $e_2 = (0, 1)^T$. Spanning alone allows redundancy; independence alone may miss directions. In a basis, every vector has a unique coordinate representation.

MAIN ROUTE · 31–34 elapsed · 3 min

LA2 · Three dependence tests — and what counts as proof

Linear Algebra tutorial guide, Week 5, p. 12, examples (i)–(iv).

Decide whether each family is linearly independent over $\mathbb{R}$. Give an explicit relation if it is dependent.

$$\text{(i)}\ \begin{pmatrix}1\\0\end{pmatrix},\ 0; \qquad \text{(ii)}\ \begin{pmatrix}1\\0\end{pmatrix},\ \begin{pmatrix}3\\0\end{pmatrix}; \qquad \text{(iii)}\ \begin{pmatrix}1\\2\end{pmatrix},\ \begin{pmatrix}1\\0\end{pmatrix},\ \begin{pmatrix}0\\2\end{pmatrix}.$$

Invent one independent family and one dependent family. When can an example settle a mathematical claim?

> **Say:** We are not trying to make every coefficient nonzero: we need at least one nonzero coefficient. In the first example, the coefficient of the zero vector is one. Notice that no pair in the third example is parallel, but all three are still dependent.

> (iii)重点纠正“两两不平行所以全体独立”。让学生说出关系，避免机械行化简。自己造例可口头让一人给一个。

All three displayed families are dependent. Writing their vectors in order, valid nontrivial relations are:

$$\text{(i)}\ 0v_1 + 1v_2 = 0; \qquad \text{(ii)}\ 3v_1 - v_2 = 0; \qquad \text{(iii)}\ v_1 - v_2 - v_3 = 0.$$

For an independent example, $\{(1,0)^T, (0,1)^T\}$: $a(1,0)^T + b(0,1)^T = 0$ forces $a = b = 0$. For a dependent example, $\{(1,1)^T, (2,2)^T\}$, since $2v_1 - v_2 = 0$.

One nontrivial relation proves dependence because dependence is an existence claim. Independence requires ruling out *every* nontrivial relation. One counterexample disproves a universal claim; one successful example does not prove it.

MAIN ROUTE · 34–37 elapsed · 3 min

LA3 · Find a basis from parameters

Linear Algebra tutorial guide, Week 5, p. 12, first basis example; real parameter domain made explicit.

Find a basis and the dimension of

$$V_1 = \{(x, y, 2x, x+y)^T : x, y \in \mathbb{R}\} \subseteq \mathbb{R}^4.$$

Give a different basis as well.

> **Say:** The ambient space has four coordinates, but this subspace has two independent parameters. Read a candidate basis from the parameters, then prove spanning and independence separately.

> 强调维数不是向量长度，也不能只数参数而不查是否冗余。这里前两坐标直接保证参数独立。第二组基一句代换即可。

Separate the free parameters:

$$\begin{pmatrix} x \\ y \\ 2x \\ x+y \end{pmatrix} = x \underbrace{\begin{pmatrix} 1 \\ 0 \\ 2 \\ 1 \end{pmatrix}}_{u} + y \underbrace{\begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}}_{v}.$$

Thus $V_1 = \operatorname{span}_{\mathbb{R}}\{u, v\}$. If $au + bv = 0$, the first two coordinates give $a = b = 0$. Therefore $\{u, v\}$ is a basis and $\dim_{\mathbb{R}} V_1 = 2$.

A different basis is $\{u+v, v\} = \{(1,1,2,2)^T, (0,1,0,1)^T\}$. It spans because $u = (u+v) - v$; it is independent because $a(u+v) + bv = au + (a+b)v = 0$ forces $a = b = 0$.

MAIN ROUTE · 37–39 elapsed · 2 min

LA4 · Find a basis from an equation

Linear Algebra tutorial guide, Week 5, p. 12, second basis example.

Find two bases and the dimension of

$$V_2 = \{(x, y, z)^T \in \mathbb{R}^3 : z = x + 2y\}.$$

> **Say:** The method is the same as the last question once we solve the constraint. An equation describing a subspace becomes a parametrisation, which becomes a spanning set, which we check for independence.

> 连接三种输入形式：约束式→参数式→span。不要把法向量误当平面内的基向量。

Use x, y as free variables:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ x + 2y \end{pmatrix} = x \underbrace{\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}}_{p} + y \underbrace{\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}}_{q}.$$

Hence $\{p, q\}$ spans V_2; the first two coordinates of $ap + bq = 0$ give $a = b = 0$. It is a basis, so $\dim V_2 = 2$. A second basis is $\{p, p + q\} = \{(1, 0, 1)^T, (1, 1, 3)^T\}$: recover $q = (p + q) - p$, and $ap + b(p + q) = 0$ forces $b = 0$, then $a = 0$.

MAIN ROUTE · 39–43 elapsed · 4 min

LA5 · Find a basis from a spanning set

Linear Algebra tutorial guide, Week 5, p. 12, third basis example; renamed V3 because the source repeats V2.

Find a basis and the dimension of

$$V_3 = \operatorname{span}_{\mathbb{R}} \left\{ \underbrace{\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}}_{a}, \underbrace{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}_{b}, \underbrace{\begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}}_{c} \right\}.$$

Can more than one subset of this spanning set be a basis?

> **Say:** Three generators do not mean dimension three. Find redundancy first. Row operations preserve relations between columns, but they generally change the column space, so take the pivot columns from the original matrix.

> 这是线代主计算题。先看出a=b+c，再展示矩阵法作为通用做法。原指南把此空间也写V2，这里改为V3，不改变任何原向量。

Spot $a = b + c$, so $c = a - b$ and $V_3 = \operatorname{span}\{a, b\}$. If $\alpha a + \beta b = 0$, its first two coordinates give $\alpha + \beta = 0$, $2\alpha + \beta = 0$, hence $\alpha = \beta = 0$. Thus $\{a, b\}$ is a basis and $\dim V_3 = 2$.

Row reduction confirms the same conclusion:

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ -1 & 1 & -2 \end{pmatrix} \xrightarrow{R_2 - 2R_1,\ R_3 + R_1} \begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 2 & -2 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}.$$

The pivot columns are columns 1 and 2; select these columns from the *original* matrix. The other two pairs, $\{a, c\}$ and $\{b, c\}$, are also bases: the relation recovers the omitted vector, and each pair is independent (their first two coordinates already force both coefficients to vanish).

LA6 · Three bases over the complex numbers

Linear Algebra tutorial guide, Week 5, p. 12, warm-up.

Write down three different bases for the complex vector space

$$V = \{(x, y, x + y)^T : x, y \in \mathbb{C}\} \subseteq \mathbb{C}^3.$$

State its dimension over $\mathbb{C}$.

> **Say:** Always say what the scalars are. Here x and y are complex and our linear combinations may have complex coefficients. Multiplying one basis vector by the nonzero scalar i gives another basis.

> 重点数域是C。三个基不等于三个向量；每个基恰有两个向量。不要转去做未交的C4作业题。

Put $u = (1, 0, 1)^T$, $v = (0, 1, 1)^T$. Then $(x, y, x + y)^T = xu + yv$, and $\alpha u + \beta v = 0$ forces $\alpha = \beta = 0$ by its first two coordinates, including for complex coefficients. Three bases are

$$B_1 = \{u, v\}, \qquad B_2 = \{u + v, u - v\}, \qquad B_3 = \{iu, v\}.$$

Explicitly, $B_2 = \{(1, 1, 2)^T, (1, -1, 0)^T\}$ and $B_3 = \{(i, 0, i)^T, (0, 1, 1)^T\}$. For B_2, recover $u = \frac{1}{2}[(u + v) + (u - v)]$, $v = \frac{1}{2}[(u + v) - (u - v)]$; a zero combination gives $\alpha + \beta = \alpha - \beta = 0$, so $\alpha = \beta = 0$. For B_3, recover $u = -i(iu)$; a zero combination gives $i\alpha = \beta = 0$. All three are bases, and $\dim_{\mathbb{C}} V = 2$.

MAIN ROUTE · 46–49 elapsed · 3 min

LA7 · What size tells us in R8

Linear Algebra tutorial guide, Week 5, p. 13; the source's condition |S|<7 is preserved, then strengthened explicitly.

Let $S \subseteq \mathbb{R}^8$. What can we say if (a) $|S| < 7$, (b) $|S| \geq 9$, or (c) $|S| = 8$? What changes at $|S| = 7$?

> **Say:** Too few cannot span; too many cannot be independent. Exactly the right number still needs a test, but then one test is enough: independent implies spanning and spanning implies independent.

> 原题确实写|S|<7，不偷偷改为<8；先回答原问，再指出真正阈值8。用集合时不要重复同一向量凑元素个数。

(a) $|S| < 7$ implies that S does not span $\mathbb{R}^8$; its independence is not determined. For example, $\{e_1\}$ is independent whereas $\{0\}$ is dependent. In fact every set with fewer than 8 vectors fails to span, so a 7-element set also cannot span.

(b) Any set with at least 9 vectors is dependent. It may span, for example $\{e_1, \ldots, e_8, 2e_1\}$, or fail to span, for example $\{e_1, 2e_1, \ldots, 9e_1\}$. For an infinite set, a dependent 9-element subset suffices.

(c) For exactly 8 vectors, independence and spanning are equivalent, and either condition means that S is a basis. Size alone does not imply either: $\{e_1, \ldots, e_8\}$ is a basis, whereas $\{e_1, 2e_1, \ldots, 8e_1\}$ is dependent and does not span.

MAIN ROUTE · 49–52 elapsed · 3 min

LA8 · Removing and adding vectors carefully

Linear Algebra tutorial guide, Week 5, p. 13, basis reduction/extension discussion; concrete examples supplied.

(a) From the spanning set $S = \{e_1, e_2, e_1 + e_2\}$ for $\mathbb{R}^2$, remove a vector to obtain a basis. Why is arbitrary removal unsafe in general?

(b) Extend the independent set $T = \{(1,1,0)^T\}$ to a basis of $\mathbb{R}^3$. What property must each newly added vector have?

> **Say:** Delete redundancy; add a genuinely missing direction. After every step, keep the relevant property: spanning when deleting, independence when adding.

> 给一条具体错误删法，避免学生记成“多就随便删、少就随便加”。若时间紧扩充只检查(a,a+b,c)=0。

(a) Delete $e_1 + e_2$; the remaining $\{e_1, e_2\}$ is a basis. In general delete a vector that lies in the span of the others, so spanning is preserved; repeat until independence holds. Arbitrary deletion can destroy spanning: deleting e_2 from $\{e_1, 2e_1, e_2\}$ leaves only the x-axis.

(b) Add $e_2 = (0,1,0)^T$, then $e_3 = (0,0,1)^T$. Each is outside the span of the vectors already chosen. If

$$a(1,1,0)^T + be_2 + ce_3 = (a, a+b, c)^T = 0,$$

then $a = b = c = 0$, so the three vectors are independent and hence form a basis of $\mathbb{R}^3$. Explicitly,

$$(x,y,z)^T = x(1,1,0)^T + (y-x)e_2 + ze_3.$$

Adding a vector w outside $\operatorname{span} T$ preserves independence: in a zero relation, a nonzero coefficient of w would express w in that span, a contradiction.

52–55 · EXIT CHECK

Two checks before you leave

1. If $g \circ f$ is injective, must f be injective? Give a one-line reason.

2. A set has eight vectors in $\mathbb{R}^8$. Is it automatically a basis? What extra fact would suffice?

1. Yes: $f(x) = f(y) \Rightarrow g(f(x)) = g(f(y)) \Rightarrow x = y$.

2. No. Proving independence or proving spanning is enough, since its size equals the dimension. For a counterexample, take $\{e_1, 2e_1, \ldots, 8e_1\}$.

> “Write down one type of question you will practise during consolidation week, and one specific mistake you will check for.”

> 最后一分钟只收一个明确学习目标；不要在最后再开逆函数新题。

Reserve answer bank

Use for questions from the room, or preparation outside the 55-minute session. Preview topics are explicitly labelled.

RESERVE / LOOKUP

P7 · Cartesian products and strict containment

Pure PS5 A1

Find integer subsets X_1, X_2, Y_1, Y_2 such that

$$(X_1 \times Y_1) \cup (X_2 \times Y_2) \subsetneq (X_1 \cup X_2) \times (Y_1 \cup Y_2).$$

> **Say:** The product on the right also includes the cross pairs.

> 备用；列出四个点最直观。

Choose $X_1 = Y_1 = \{1\}$, $X_2 = Y_2 = \{2\}$. The left side is $\{(1,1),(2,2)\}$; the right side is $\{(1,1),(1,2),(2,1),(2,2)\}$. Every left-side element lies on the right, and $(1,2)$ lies only on the right. This proves strict containment.

RESERVE / LOOKUP

P8 · A bijection from integers to multiples of three

Pure PS5 A3

Construct and prove a bijection $f : \mathbb{Z} \to 3\mathbb{Z}$.

> **Say:** The codomain matters: this is surjective onto three Z, but not onto all of Z.

> 备用；同一公式改余域会改变满射性。

Define $f(n) = 3n$, which belongs to $3\mathbb{Z}$. If $f(m) = f(n)$, then $3m = 3n$, so $m = n$: injective. Every $y \in 3\mathbb{Z}$ has the form $y = 3n$ for an integer n, and then $f(n) = y$: surjective. Thus f is bijective.

RESERVE / LOOKUP

P9 · Preimages respect inclusion

Pure PS5 A4

Let $f : X \to Y$ and $U, V \subseteq Y$. Prove or disprove: $U \subseteq V$ implies $f^{-1}[U] \subseteq f^{-1}[V]$.

> **Say:** Translate membership using the definition; apply the given inclusion; translate back.

> 备用；强调不要求 f 单射或满射。

The statement is true. Take $x \in f^{-1}[U]$. By definition $f(x) \in U$. Since $U \subseteq V$, we have $f(x) \in V$, hence $x \in f^{-1}[V]$. This proves the claimed inclusion.

RESERVE / LOOKUP

P10 · Advance material: two self-inverse maps

Pure PS5 A7(a–d); inverse functions / reserve

Let $X = \mathbb{R} \setminus \{-1, 0, 1\}$, $f(x) = (x+1)/(x-1)$, and $g(x) = 1/x$. Check both maps are $X \to X$, compute $g \circ f$ and $f \circ g$, prove $f = f^{-1}$, $g = g^{-1}$, and find $(f \circ g)^{-1}$.

> **Say:** First check the excluded values, then substitute carefully. The inverse of a composition reverses the order.

> 提前内容 / 备用；周五 13:00 才讲逆函数，早上不默认已学。注意 f∘g 分母是 1−x。

For $x \in X$, both denominators are nonzero. If $f(x) = 0$ or -1, then respectively $x = -1$ or $x = 0$, both excluded. The equation $f(x) = 1$ gives $1 = -1$, impossible. For g, the values $1, -1$ require $x = 1, -1$, and $g(x) = 0$ is impossible. Thus both outputs belong to X.

Direct substitution gives

$$(g \circ f)(x) = \frac{x-1}{x+1}, \qquad (f \circ g)(x) = \frac{x+1}{1-x}.$$

Also

$$f(f(x)) = \frac{(x+1)+(x-1)}{(x+1)-(x-1)} = x, \qquad g(g(x)) = x.$$

Since each is a self-map and composing it with itself gives the identity, each is its own inverse. Therefore

$$(f \circ g)^{-1} = g^{-1} \circ f^{-1} = g \circ f, \qquad (f \circ g)^{-1}(x) = \frac{x-1}{x+1}.$$

Checking directly, if $h(x) = (x+1)/(1-x)$, $k(x) = (x-1)/(x+1)$, then both $h(k(x)) = x$ and $k(h(x)) = x$.

RESERVE / LOOKUP

P11 · Advance material: a one-sided inverse

Pure PS5 A8(a–c); inverse functions / reserve

On $\mathbb{Z}$, let $f(x) = 2x$, and let $g(x) = x/2$ for even x, $g(x) = (x+1)/2$ for odd x. Show neither map is bijective; show $g \circ f$ is the identity; decide whether $f \circ g$ is the identity or bijective.

> **Say:** Undoing in one order does not automatically undo in the other order.

> 提前内容 / 备用；反例同时支撑 B10、B11。

The map f misses every odd integer, so is not surjective. Also $g(1) = g(2) = 1$ with $1 \neq 2$, so g is not injective. Neither is bijective. For every integer x, $2x$ is even and $g(f(x)) = g(2x) = x$. Thus $g \circ f$ is the identity, hence bijective. In the reverse order,

$$(f \circ g)(x) = \begin{cases} x, & x \text{ even}, \\ x+1, & x \text{ odd}. \end{cases}$$

This is not the identity, since $1 \mapsto 2$; it is not injective, since $1, 2 \mapsto 2$; and it is not surjective, since its image consists only of even integers.

RESERVE / LOOKUP

P12 · Advance material: translation and scaling

Pure PS5 A12(a–b); inverse functions / reserve

For fixed $a \in \mathbb{R}$, prove $T_a(x) = x + a$ is a bijection $\mathbb{R} \to \mathbb{R}$ by finding its inverse. For $a \neq 0$, do the same for $M_a(x) = ax$.

> **Say:** Find a candidate, state its domain, and check both compositions.

> 提前内容 / 备用；缩放必须 a≠0。

Set $S_a(y) = y - a$. Then $S_a(T_a(x)) = (x + a) - a = x$ and $T_a(S_a(y)) = (y - a) + a = y$, so $T_a^{-1} = S_a$. For $a \neq 0$, set $N_a(y) = y/a$. Then $N_a(M_a(x)) = ax/a = x$, $M_a(N_a(y)) = a(y/a) = y$, so $M_a^{-1} = N_a$. Both are bijections because they have two-sided inverses.

RESERVE / LOOKUP

P13 · A rational bijection onto positive reals

Pure PS5 B1(a–c)

For $f:(0,1)\to(0,\infty)$, $f(x)=(1-x)/x$, prove injectivity, surjectivity, and bijectivity.

> **Say:** Solving for x is only half the surjectivity proof: check that x lies in the domain.

> 备用；可作为 B3 的替代题，不能只解方程不检查区间。

For $0<x<1$, both numerator and denominator are positive, so the map has the stated codomain. If $(1-x_1)/x_1=(1-x_2)/x_2$, cross-multiplication gives $x_2-x_1x_2=x_1-x_1x_2$, hence $x_1=x_2$: injective. Given arbitrary $y>0$, choose $x=1/(1+y)$. Since $1+y>1$, $x\in(0,1)$; moreover

$$f(x)=\frac{1-1/(1+y)}{1/(1+y)}=y.$$

Thus f is surjective and hence bijective.

RESERVE / LOOKUP

P14 · Advance material: a bijection onto an open interval

Pure PS5 B2(a–c); inverse functions / reserve

For $f:\mathbb{R}\to(-1,1)$, $f(x)=x/(1+|x|)$, verify its range lies in $(-1,1)$, prove surjectivity, and find an inverse.

> **Say:** The denominator of the inverse is one minus the absolute value. It is positive precisely because the target lies in the open interval.

> 提前内容 / 备用；不能漏绝对值或把 1−|y| 写成 1+|y|。

We have $|f(x)|=|x|/(1+|x|)<1$. For $y\in(-1,1)$, put $g(y)=y/(1-|y|)$; its denominator is positive. Then

$$|g(y)|=\frac{|y|}{1-|y|},\qquad f(g(y))=\frac{y/(1-|y|)}{1+|y|/(1-|y|)}=y.$$

This proves surjectivity. Also $1-|f(x)|=1/(1+|x|)$, so

$$g(f(x))=\frac{x/(1+|x|)}{1-|x|/(1+|x|)}=x.$$

Both compositions are identities; hence $g=f^{-1}$.

RESERVE / LOOKUP

P15 · An integer map with a constructive preimage

Pure PS5 B4(a–c)

Let $f : \mathbb{Z}^3 \to \mathbb{Z}^2$, $f(m, n, p) = (n - p^2, m + n^2)$. Show it is not injective, is surjective, and determine whether it is bijective.

> **Say:** For surjectivity, choose the free parameter to make life easy: take p equal to zero.

> 备用；解出一个合法原像即可，无需枚举所有。

The distinct inputs $(0, 0, 0)$ and $(-1, 1, 1)$ both map to $(0, 0)$, so f is not injective. Given arbitrary $(u, v) \in \mathbb{Z}^2$, choose $p = 0$, $n = u$, $m = v - u^2$, all integers. Then

$$f(v - u^2, u, 0) = (u, v - u^2 + u^2) = (u, v).$$

Thus f is surjective. It is not bijective because it is not injective.

RESERVE / LOOKUP

P16 · A product of bijections

Pure PS5 B5

If $f : X \to \mathbb{Z}_{>0}$ and $g : Y \to \mathbb{Z}_{>0}$ are bijections, prove $h : X \times Y \to \mathbb{Z}_{>0}^2$, $h(x, y) = (f(x), g(y))$, is bijective.

> **Say:** Handle the two coordinates independently, then combine the witnesses into one ordered pair.

> 备用；有序对相等等价于对应坐标相等。

If $h(x_1, y_1) = h(x_2, y_2)$, equality of coordinates gives $f(x_1) = f(x_2)$ and $g(y_1) = g(y_2)$. Their injectivity gives $x_1 = x_2$, $y_1 = y_2$, so h is injective. For arbitrary $(u, v) \in \mathbb{Z}_{>0}^2$, surjectivity of f, g gives $x \in X, y \in Y$ with $f(x) = u, g(y) = v$. Then $h(x, y) = (u, v)$, so h is surjective and therefore bijective.

RESERVE / LOOKUP

P17 · Hand-in question: guidance only

Pure PS5 B8(a–b); due Thu 29 Oct 2026, 13:00; peer marking due Thu 5 Nov, 13:00

Suppose $f : X \to Y$, $U \subseteq X$, and $V_1, V_2 \subseteq Y$.

(a) Show $f^{-1}[V_1 \cup V_2] = f^{-1}[V_1] \cup f^{-1}[V_2]$.

(b) Show $U \subseteq f^{-1}[f[U]]$, and that equality holds when f is injective.

> **Say:** Use the definitions and label your two inclusions. For the stronger conclusion, point to the exact line where injectivity is needed.

> 仅提示，不完整解答；题目字段只有题干。不得将教师解答稿直接发学生。

Hints only — no complete solution. For (a), start with an arbitrary input and translate preimage membership into a statement about its output. Which logical connective corresponds to union? For (b), prove the requested inclusion first; then treat the reverse inclusion as a separate argument under injectivity. Identify exactly where that extra assumption becomes necessary. Review A4, A6, and the witness notation used in B7.

Students should write their own proof. This is the current peer-marked hand-in question; do not project or circulate a complete model answer before its deadline.

RESERVE / LOOKUP

P18 · Advance material: uniqueness of inverse

Pure PS5 B9; inverse functions / reserve

If $g, h : Y \to X$ are both inverses of $f : X \to Y$, prove $g = h$.

> **Say:** Insert one identity, then use the other identity.

> 提前内容 / 备用；写清作用域 Y 与 X，不能把两个 identity 混用。

For arbitrary $y \in Y$, since $f \circ h = \mathrm{id}_Y$ and $g \circ f = \mathrm{id}_X$,

$$g(y) = g(f(h(y))) = h(y).$$

Thus the two functions agree at every point of their common domain, so $g = h$.

RESERVE / LOOKUP

P19 · Advance material: completing a one-sided inverse

Pure PS5 B10(a–b); inverse functions / reserve

Suppose $f : X \to Y$, $g : Y \to X$, and $g \circ f = \mathrm{id}_X$. Prove $f \circ g = \mathrm{id}_Y$ if (a) g is injective; or (b) f is surjective.

> **Say:** In part a, cancel using injectivity. In part b, represent the arbitrary target as f of something.

> 提前内容 / 备用；与 A8 对照，单边逆需要额外条件。

(a) Take $y \in Y$. Applying the given identity to $g(y) \in X$ gives $g(f(g(y))) = g(y)$. Since g is injective, the two inputs $f(g(y))$ and y must be equal. Thus $f \circ g = \mathrm{id}_Y$.

(b) Take $y \in Y$. Surjectivity provides $x \in X$ with $f(x) = y$. Then

$$f(g(y)) = f(g(f(x))) = f(x) = y.$$

Again $f \circ g = \mathrm{id}_Y$.

RESERVE / LOOKUP

P20 · Advance material: what a bijective composition implies

Pure PS5 B11(a–d); inverse functions / reserve

Let $f : X \to Y$, $g : Y \to Z$, and suppose $g \circ f$ is bijective. Prove (a) f is injective; (b) g is surjective; (c) if f is invertible then $(g \circ f) \circ f^{-1} = g$, and deduce g is bijective; (d) if g is invertible then $g^{-1} \circ (g \circ f) = f$, and deduce f is bijective.

> **Say:** Information lost by the first map cannot be recovered by the second. A missing final target cannot be fixed by the first map.

> 提前内容 / 备用；a,b 可先讲，但整题不列主线。

(a) If $f(x_1) = f(x_2)$, applying g gives $(g \circ f)(x_1) = (g \circ f)(x_2)$; injectivity of the composition gives $x_1 = x_2$.

(b) For each $z \in Z$, surjectivity of the composition supplies $x \in X$ with $g(f(x)) = z$. The point $y = f(x) \in Y$ is a preimage under g.

(c) Associativity and $f \circ f^{-1} = \mathrm{id}_Y$ give $(g \circ f) \circ f^{-1} = g$. Both $g \circ f$ and f^{-1} are bijective, so their composition g is bijective.

(d) Similarly, $g^{-1} \circ (g \circ f) = (g^{-1} \circ g) \circ f = f$; the two factors on the left are bijective, so f is bijective.

Without the extra hypotheses, the factors need not both be bijective: A8 is a counterexample.

RESERVE / LOOKUP

P21 · Advance material: a fractional-linear inverse

Pure PS5 B12(a–d); inverse functions / reserve

For $a, b > 0$, let $X_a = \mathbb{R} \setminus \{a\}$, $X_b = \mathbb{R} \setminus \{b\}$, and $f_{a,b} : X_a \to X_b$, $f_{a,b}(x) = (bx + 1)/(x - a)$. Verify the map, find and verify its inverse, and disprove the claim $f_{b,c} \circ f_{a,b} = f_{a,c}$ for all positive a, b, c.

> **Say:** A proposed inverse must land in the right set. To disprove a statement about all parameters, choose the simplest legal parameters and one legal input.

> 提前内容 / 备用；原解 B12(d) a,b,c 标记混乱，本稿重新取全为 1 并验证。

(a) For $x \neq a$, the denominator is nonzero. Moreover

$$f_{a,b}(x) - b = \frac{1 + ab}{x - a} \neq 0,$$

since $1 + ab > 0$; thus the output lies in X_b.

(b) Solving $y = (bx + 1)/(x - a)$ gives $x = (ay + 1)/(y - b)$. Define $g_{a,b} : X_b \to X_a$ by this formula. Its denominator is nonzero and $g_{a,b}(y) - a = (1 + ab)/(y - b) \neq 0$, so it has the required domain and codomain.

(c) Direct substitution gives

$$g_{a,b}(f_{a,b}(x)) = \frac{x(ab + 1)}{ab + 1} = x, \qquad f_{a,b}(g_{a,b}(y)) = \frac{y(ab + 1)}{ab + 1} = y.$$

Therefore $g_{a,b} = f_{a,b}^{-1} = f_{b,a}$.

(d) Choose $a = b = c = 1$ and $x = 0 \in X_1$. Then $f_{1,1}(0) = -1$ and $f_{1,1}(-1) = 0$. Consequently

$$(f_{1,1} \circ f_{1,1})(0) = 0 \neq -1 = f_{1,1}(0).$$

This valid instance disproves the universal claim. It avoids the inconsistent parameter labels in the supplied model solution.

RESERVE / LOOKUP

P22 · Advance material: reverse the order for inverse composition

Pure PS5 B13; inverse functions / reserve

For bijections $f : X \to Y$ and $g : Y \to Z$, prove $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

> **Say:** Undo the last operation first. The domains also force the reversed order.

> 提前内容 / 备用；可用 X→Y→Z 的箭头帮助检查公式能否复合。

Define $h = f^{-1} \circ g^{-1} : Z \to X$. Associativity gives

$$h \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ f = \mathrm{id}_X.$$

In the other order,

$$(g \circ f) \circ h = g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ g^{-1} = \mathrm{id}_Z.$$

Hence h is the inverse of $g \circ f$, proving the formula.

RESERVE / LOOKUP

P23 · Reserve: encode a finite-support sequence with primes

Pure PS5 B14; later material / reserve

Let X consist of sequences $(a_1, a_2, \ldots)$ of nonnegative integers with only finitely many nonzero terms. Construct a bijection $X \to \mathbb{Z}_{>0}$.

> **Say:** The prime is the position label; the exponent is the value stored at that position. Finite support is what makes the product an ordinary integer.

> 备用 / 后续内容；起始指标 i=1，单独处理 n=1 与空乘积；不能把任意无限序列也映到整数。

Let p_i be the i-th prime, starting with $p_1 = 2$, and define

$$F((a_i)_{i \geq 1}) = \prod_{i : a_i \neq 0} p_i^{a_i}.$$

This is a finite product by the definition of X, hence a positive integer. The empty product is 1, so the all-zero sequence maps to 1.

Injective. If two such products agree, choose N beyond the support of both sequences. Unique prime factorisation of the equal products $\prod_{i=1}^{N} p_i^{a_i} = \prod_{i=1}^{N} p_i^{b_i}$ gives $a_i = b_i$ for all $1 \leq i \leq N$. Beyond N both sequences are zero, so they are equal.

Surjective. Every $n > 1$ has a prime factorisation with finitely many nonzero exponents. Put those exponents at the corresponding prime indices and put zero elsewhere; this gives an element of X mapped to n. The integer 1 is attained by the all-zero sequence. Thus F is bijective.

RESERVE / LOOKUP

P24 · Challenge: images of set differences

Pure PS5 C1, C2; reserve

C1. If $f : X \to Y$ is not injective, find $A, B \subseteq X$ such that $f[A \setminus B] \not\subseteq f[A] \setminus f[B]$.

C2. If f is injective, prove $f[A \setminus B] \subseteq f[A] \setminus f[B]$ for all $A, B \subseteq X$.

> **Say:** For set difference, prove membership in the first set and non-membership in the second.

> 挑战备用；本题无官方答案，本稿按定义独立证明。

C1. Choose distinct $a, b \in X$ with $f(a) = f(b)$, and put $A = \{a\}$, $B = \{b\}$. Then $A \setminus B = A$, so $f[A \setminus B] = \{f(a)\}$, while $f[A] \setminus f[B] = \varnothing$. A nonempty set is not a subset of the empty set.

C2. Let $y \in f[A \setminus B]$. Then $y = f(a)$ for some $a \in A \setminus B$, so $y \in f[A]$. If $y \in f[B]$, choose $b \in B$ with $f(b) = y = f(a)$. Injectivity gives $b = a$, contradicting $a \notin B$. Therefore $y \notin f[B]$, so $y \in f[A] \setminus f[B]$, proving the inclusion.

RESERVE / LOOKUP

P25 · Challenge: restrict the codomain or the domain

Pure PS5 C3(a–b); reserve

Let X, Y be finite sets and $f : X \to Y$. (a) If f is injective, find $B \subseteq Y$ such that the same rule gives a bijection $X \to B$. (b) If f is surjective, find $A \subseteq X$ such that $f|_A : A \to Y$ is bijective.

> **Say:** For an injection, discard unreachable targets. For a surjection, keep exactly one representative of each fibre.

> 挑战备用；这里余域仍是 Y，定义域缩到 A；空集情形已覆盖。

(a) Set $B = f[X]$. The same rule is injective because f is, and is surjective onto B by the definition of image; hence it is bijective. Finiteness is not needed for this part.

(b) If $Y = \varnothing$, take $A = \varnothing$. Otherwise list its distinct elements as $y_1, \ldots, y_n$. Surjectivity lets us choose $x_i \in X$ with $f(x_i) = y_i$. Since the y_i are distinct, the chosen x_i are distinct. Put $A = \{x_1, \ldots, x_n\}$. The restriction attains every y_i, so is surjective. If $f(x_i) = f(x_j)$, then $y_i = y_j$, so $i = j$ and $x_i = x_j$; it is injective. Thus it is bijective.

RESERVE / LOOKUP

P26 · Advance challenge: shift a sequence inside an interval

Pure PS5 C4 (all four requests; original lettering duplicated); inverse functions / reserve

Define $f : [0,1) \to (0,1)$ by

$$f(x) = \begin{cases} 1 - \dfrac{1}{n+1}, & x = 1 - \dfrac{1}{n} \text{ for some } n \in \mathbb{Z}_{>0}, \\ x, & \text{otherwise.} \end{cases}$$

Check that this is a map into $(0,1)$, show surjectivity, construct g with $g \circ f = \mathrm{id}_{[0,1)}$, and verify $f \circ g = \mathrm{id}_{(0,1)}$.

> **Say:** Move zero to one-half, one-half to two-thirds, and so on; leave every other point fixed. The inverse shifts that same chain backwards.

> 挑战 / 提前内容；特殊点是 0,1/2,2/3,3/4,…，不是 1/n。原题四个要求的 a 标号重复，已按四步骤完整处理。

Write $s_n = 1 - 1/n$ for $n \geq 1$, and $S = \{s_n : n \geq 1\}$. The sequence is strictly increasing, so each point in S has a unique index. Also $s_1 = 0$, and the rule is $f(s_n) = s_{n+1}$; all points outside S stay fixed.

Well-defined range. Every s_{n+1} lies in $(0,1)$. If $x \in [0,1) \setminus S$, then $x \neq 0$, so its fixed image also lies in $(0,1)$.

Surjectivity and candidate inverse. Each special target s_n, $n \geq 2$, is hit by s_{n-1}. A target outside S is hit by itself. Thus define

$$g(y) = \begin{cases} s_{n-1} = 1 - \dfrac{1}{n-1}, & y = s_n \text{ for some } n \geq 2, \\ y, & y \notin S. \end{cases}$$

This is well-defined on $(0,1)$ and takes values in $[0,1)$.

Both compositions. For $x = s_n$, $g(f(x)) = g(s_{n+1}) = s_n = x$; outside S, both operations fix x. Hence $g \circ f = \mathrm{id}_{[0,1)}$. For $y = s_n$, $n \geq 2$, $f(g(y)) = f(s_{n-1}) = s_n = y$; outside S, both fix y. Hence $f \circ g = \mathrm{id}_{(0,1)}$, and $g = f^{-1}$.

RESERVE / LOOKUP

LA9 · Reserve: intersecting bases is not intersecting spans

Linear Algebra tutorial guide, Week 5, p. 13, challenging question and footnote 7; missing y parameter in source footnote corrected.

Let V, W be subspaces of $\mathbb{R}^3$, with respective bases B_V, B_W. Must $B_V \cap B_W$ be a basis of $V \cap W$? Prove it or give a counterexample.

> **Say:** The intersection of two chosen lists of vectors only sees literally identical vectors. The intersection of the subspaces sees every vector you can build in both spaces.

> 备用2分钟或课后阅读。原脚注V只写x∈R漏了y∈R，已补为x,y∈R，使其与给定基一致。

No. Take

$$V = \{(x, y, 0)^T : x, y \in \mathbb{R}\}, \qquad W = \{(0, y, z)^T : y, z \in \mathbb{R}\},$$

$$B_V = \{(1, 0, 0)^T, (0, 1, 0)^T\}, \qquad B_W = \{(0, 2, 0)^T, (0, 0, 1)^T\}.$$

These are bases of the indicated planes. Their set intersection is empty, whereas

$$V \cap W = \{(0, y, 0)^T : y \in \mathbb{R}\} \neq \{0\}.$$

The empty set spans only $\{0\}$, so it is not a basis of this intersection. The shared direction is present in both spans even though the chosen basis vectors are different. Any $B_V \cap B_W$ is independent, being a subset of a basis, but may fail to span $V \cap W$.

RESERVE / LOOKUP

LA10 · Consolidation checkpoint

Linear Algebra tutorial guide, Week 5, p. 13, End and hand-in boundary.

Choose one concrete consolidation goal: (i) find bases from all three descriptions of a subspace; (ii) justify independence using a zero relation; or (iii) distinguish spanning, independence and dimension. Write down one question you still have.

> **Say:** Choose one thing you will practise during consolidation week. When you write that something is a basis, make sure your argument explains both spanning and independence.

> 原指南提示下周consolidation无tutorial、week6无hand-in，未独立核对实际日历时不要额外承诺课程安排。Week7未交题不讲答案、不复制给学生；最多提示“先按参数展开，再检查独立；普遍命题可先想反例”。

A useful target is specific and checkable: “I will solve one parameter, one equation and one spanning-set example, and give both parts of the basis proof without looking at the solutions.”

RESERVE / LOOKUP

LA5.2 · Real independence tests

Linear Algebra Problem Sheet 5, Question 5.2 (local LA-questions.txt, pp. 25–29).

Decide whether each set is linearly independent over $\mathbb{R}$. Explain.

$$\text{(a)}\ \left\{\begin{pmatrix}2\\1\end{pmatrix},\begin{pmatrix}-14\\-7\end{pmatrix}\right\};\quad \text{(b)}\ \left\{\begin{pmatrix}1\\-1\end{pmatrix},\begin{pmatrix}-1\\2\end{pmatrix}\right\};\quad \text{(c)}\ \left\{\begin{pmatrix}1\\-1\\1\end{pmatrix},\begin{pmatrix}1\\1\\1\end{pmatrix},\begin{pmatrix}2\\0\\2\end{pmatrix}\right\}.$$

> **Say:** Look for a relation before doing elimination. If two real plane vectors are not scalar multiples, their determinant is nonzero.

> 三小题快速辨认；5.2(c)三向量虽无零向量、无成比例对，仍可相关。

(a) Dependent: $v_2 = -7v_1$, so $7v_1 + v_2 = 0$ is nontrivial. **(b) Independent:** the column matrix has determinant $1 \cdot 2 - (-1)(-1) = 1 \neq 0$. **(c) Dependent:** $v_3 = v_1 + v_2$, so $v_1 + v_2 - v_3 = 0$ is nontrivial.

RESERVE / LOOKUP

LA5.3 · Complex independence tests

Linear Algebra Problem Sheet 5, Question 5.3 (local LA-questions.txt, pp. 25–29).

Decide whether each set is linearly independent over $\mathbb{C}$. Vectors are listed in order.

$$\text{(a)}\ \left\{\begin{pmatrix}1-i\\2\end{pmatrix},\begin{pmatrix}3\\1+i\end{pmatrix},\begin{pmatrix}-1-2i\\3-i\end{pmatrix}\right\};$$

$$\text{(b)}\ \left\{\begin{pmatrix}1\\-i\\i\end{pmatrix},\begin{pmatrix}i\\1\\-1\end{pmatrix}\right\};\qquad \text{(c)}\ \left\{\begin{pmatrix}i\\-1\\0\end{pmatrix},\begin{pmatrix}1\\2i\\1\end{pmatrix},\begin{pmatrix}2\\0\\2\end{pmatrix}\right\}.$$

> **Say:** Over the complex numbers, i is a perfectly valid coefficient. In part (a), dimension already decides the question; the explicit relation lets us check the arithmetic.

> (b)不要因不是实倍数就判独立；(c)保留i²=-1的符号。三向量的所有坐标已按原题核对。

(a) Dependent: there are three vectors in the two-dimensional complex space $\mathbb{C}^2$. Explicitly, $v_3 = 2v_1 - v_2$, giving $2v_1 - v_2 - v_3 = 0$.

(b) Dependent over $\mathbb{C}$: $v_2 = iv_1$. The allowed scalar i is essential here.

(c) Independent:

$$\det\begin{pmatrix}i & 1 & 2\\-1 & 2i & 0\\0 & 1 & 2\end{pmatrix} = i(4i) - (-2) + 2(-1) = -4 \neq 0.$$

Hence the homogeneous system has only the zero solution.

RESERVE / LOOKUP

LA5.4 · Bases for homogeneous solution spaces

Linear Algebra Problem Sheet 5, Question 5.4 (local LA-questions.txt, pp. 25–29).

For each matrix, find a basis for $S(A,0)=\{x : Ax=0\}$, using the stated scalar field.

$$\text{(a) } A=\begin{pmatrix}1 & i\\ 2i & -2\end{pmatrix}\in M_2(\mathbb{C}); \quad \text{(b) } A=\begin{pmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{pmatrix}\in M_{2,3}(\mathbb{R});$$

$$\text{(c) } A=\begin{pmatrix}0 & -1 & 2\\ 1 & 3 & -1\\ 2 & -2 & 1\end{pmatrix}\in M_3(\mathbb{R}).$$

> **Say:** A nullspace basis comes from free variables in Ax equals zero. Do not confuse it with the pivot columns, which give a basis of the column space. The zero space has an empty basis.

> 5.4(a)数域C、(b)(c)R。重点区分零空间基与列空间基，(c)答案空集不是{0}。有理数消元独立验证(b)(c)。

(a) Row 2 is $2i$ times row 1, so $x_1+ix_2=0$. Therefore

$$S(A,0)=\{t(-i,1)^T : t\in\mathbb{C}\}, \qquad B=\{(-i,1)^T\}.$$

The vector is nonzero, hence independent; it spans by the displayed parametrisation.

(b) The reduced matrix is

$$\begin{pmatrix}1 & 0 & -1\\ 0 & 1 & 2\end{pmatrix}, \quad x_1=x_3, \quad x_2=-2x_3.$$

Thus $S(A,0)=\{t(1,-2,1)^T : t\in\mathbb{R}\}$, with basis $\{(1,-2,1)^T\}$.

(c) $\det A=3-16=-13\neq 0$, so $S(A,0)=\{0\}$. Its basis is the **empty set** and its dimension is zero. The set $\{0\}$ is not a basis, since it is dependent.

RESERVE / LOOKUP

LA5.5 · A sign change changes independence

Linear Algebra Problem Sheet 5, Question 5.5 (local LA-questions.txt, pp. 25–29).

Use determinants to decide whether $v_1 = (1, 0, -i)^T$, $v_2 = (1, 1, 1)^T$, $v_3 = (i, 0, 1)^T$ are linearly independent over $\mathbb{C}$. Repeat after replacing v_3 by $(i, 0, -1)^T$.

> **Say:** Choose the row with two zeros to keep the determinant short, and calculate i times minus i carefully.

> 先按题目要求用行列式，再给显然倍数关系作交叉检查。

Expanding along the second row gives

$$\det \begin{pmatrix} 1 & 1 & i \\ 0 & 1 & 0 \\ -i & 1 & 1 \end{pmatrix} = 1 - i(-i) = 0.$$

Thus the original vectors are dependent; indeed $v_3 = iv_1$. After the change,

$$\det \begin{pmatrix} 1 & 1 & i \\ 0 & 1 & 0 \\ -i & 1 & -1 \end{pmatrix} = -1 - i(-i) = -2 \neq 0,$$

so the new three vectors are independent.

RESERVE / LOOKUP

LA5.6 · Extract a basis and express the omitted columns

Linear Algebra Problem Sheet 5, Question 5.6 (local LA-questions.txt, pp. 25–29).

In each case the columns of the displayed matrix form the set $S = \{v_1, \dots, v_k\}$. Let $V = \operatorname{span}_{\mathbb{R}} S$. Choose a basis from S, and express every omitted vector as a linear combination of your chosen basis.

$$\text{(a) } A = \begin{pmatrix} 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 5 \\ 1 & -5 & -3 & -2 \end{pmatrix}; \quad \text{(b) } A = \begin{pmatrix} 1 & 1 & 5 & 0 & 1 \\ 1 & 2 & 7 & 1 & 1 \\ 1 & 2 & 7 & 1 & 2 \end{pmatrix}.$$

> **Say:** The pivot positions tell us which original vectors to keep. The nonpivot columns tell us the coefficients expressing the vectors we leave out.

> 重点备选计算题。原题所有3×4、3×5坐标及RREF已精确分数消元核验。别把rref的标准基抄成“从S选出的基”。

(a) Exact row reduction gives

$$\operatorname{rref} A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

The original columns v_1, v_2, v_3 are independent and span V; their determinant is $18 \neq 0$. Thus

$$B = \{(1,1,1)^T, (1,1,-5)^T, (1,4,-3)^T\}, \quad v_4 = v_1 + v_3, \quad V = \mathbb{R}^3.$$

(b) Exact row reduction gives

$$\operatorname{rref} A = \begin{pmatrix} 1 & 0 & 3 & -1 & 0 \\ 0 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Pivot columns are 1, 2 and 5, so choose the corresponding original columns:

$$B = \{v_1, v_2, v_5\} = \{(1,1,1)^T, (1,2,2)^T, (1,1,2)^T\}.$$

Their determinant is $1 \neq 0$, and the omitted columns satisfy

$$v_3 = 3v_1 + 2v_2, \qquad v_4 = -v_1 + v_2.$$

Hence again $V = \mathbb{R}^3$ and $\dim V = 3$. Row operations preserve linear relations among columns; basis vectors must be taken from the original matrix.

RESERVE / LOOKUP

LA5.7 · Independence is a trivial kernel

Linear Algebra Problem Sheet 5, Question 5.7 (local LA-questions.txt, pp. 25–29).

Let $A \in M_{m,n}(\mathbb{R})$, with columns $a_1, \ldots, a_n$. Prove that these columns are linearly independent if and only if $S(A, 0) = \{0\}$.

> **Say:** The unknowns in Ax equals zero are exactly the coefficients of a linear relation between the columns.

> 这是5.3–5.6算法背后的定义桥梁；两方向都只用Ax列展开。

For $x = (x_1, \ldots, x_n)^T$, matrix multiplication gives

$$Ax = x_1 a_1 + \cdots + x_n a_n.$$

If the columns are independent, $Ax = 0$ forces every $x_j = 0$, so $S(A, 0) = \{0\}$. Conversely, if $S(A, 0) = \{0\}$, any relation $\sum_j x_j a_j = 0$ gives $Ax = 0$, hence $x = 0$. Thus all coefficients vanish, proving independence.

RESERVE / LOOKUP

LA5.8 · Subsets of independent and dependent sets

Linear Algebra Problem Sheet 5, Question 5.8 (local LA-questions.txt, pp. 25–29).

(a) Prove that every nonempty subset of a linearly independent set $S = \{v_1, \ldots, v_k\}$ is linearly independent. (b) Does the same statement hold with “dependent” in place of “independent”?

> **Say:** Removing vectors cannot create a new dependence. It can remove the dependence that was already there.

> 反例写清两元素是e1与2e1，系数2、−1；不是同一向量重复计数。

(a) Take any zero linear combination of vectors from the subset. Assign coefficient zero to every vector of S outside the subset. This gives a zero relation among all vectors of S; independence forces all coefficients to vanish, including the original coefficients. Hence the subset is independent. (The empty subset is also independent.)

(b) No. In $\mathbb{R}^2$, $S = \{e_1, 2e_1\}$ is dependent since $2e_1 - (2e_1) = 0$ uses coefficients $2, -1$, but $\{e_1\}$ is independent.

RESERVE / LOOKUP

LA5.9 · Rational independence forces irrational entries

Linear Algebra Problem Sheet 5, Question 5.9 (local LA-questions.txt, pp. 25–29).

Suppose the listed family $1, s_1, \ldots, s_n$ of real numbers is linearly independent over $\mathbb{Q}$. Show that each s_j is irrational.

> **Say:** A rational value would give us a rational linear relation with 1. The denominator is nonzero, so the relation really is nontrivial.

> 按原题意把{1,s1,…,sn}明确作所列向量族。若逐字按集合且允许重复列出1，原陈述有歧义；独立性应作用于所列项（或要求这n+1个元素互异）。

Fix j. If $s_j = p/q$ for integers p, q with $q \neq 0$, then

$$p \cdot 1 - q \cdot s_j = 0$$

is a rational linear relation in the displayed family, with all other coefficients zero. Its coefficient $-q$ is nonzero, contradicting independence. Therefore $s_j \notin \mathbb{Q}$ for every j.

This uses independence over $\mathbb{Q}$; changing the scalar field changes the claim.

RESERVE / LOOKUP

LA5.10 · Changing the scalar field doubles dimension

Linear Algebra Problem Sheet 5, Question 5.10 (local LA-questions.txt, pp. 25–29).

Let e_1, e_2 be the standard complex basis of $\mathbb{C}^2$.

(a) Show that $\{e_1, ie_1, e_2, ie_2\}$ is a real basis of $\mathbb{C}^2$. (b) Compare $\dim_{\mathbb{R}} \mathbb{C}^2$ and $\dim_{\mathbb{C}} \mathbb{C}^2$. (c) If $v_1, \ldots, v_n$ is a complex basis of a finite-dimensional complex vector space V, prove that $v_1, iv_1, \ldots, v_n, iv_n$ is a real basis. (d) Relate the two dimensions of V.

> **Say:** The vectors do not change; the allowed coefficients do. With real coefficients, the imaginary direction needs its own basis vector.

> 修正原解把一般V写成C^n、求和中vi/vk混用的明显笔误。只说V本身，不假设它字面等于C^n。

(a) Any $(z_1, z_2)^T$, with $z_j = a_j + ib_j$ and real a_j, b_j, equals

$$a_1 e_1 + b_1(ie_1) + a_2 e_2 + b_2(ie_2).$$

This proves real spanning. If this expression is zero, each coordinate gives $a_j + ib_j = 0$; equality of real and imaginary parts gives $a_j = b_j = 0$. Thus it is a real basis.

(b) $\dim_{\mathbb{R}} \mathbb{C}^2 = 4 = 2 \dim_{\mathbb{C}} \mathbb{C}^2$.

(c) Every $v \in V$ has an expansion

$$v = \sum_{j=1}^{n} (a_j + ib_j) v_j = \sum_{j=1}^{n} \big(a_j v_j + b_j (iv_j)\big), \qquad a_j, b_j \in \mathbb{R},$$

so the proposed family spans V over $\mathbb{R}$. A zero real combination becomes $\sum_j (a_j + ib_j) v_j = 0$; complex independence gives $a_j + ib_j = 0$, hence all real coefficients vanish. This proves real independence.

(d) The real basis has $2n$ vectors, so $\dim_{\mathbb{R}} V = 2 \dim_{\mathbb{C}} V$.

RESERVE / LOOKUP

LA5.11 · A product of vector spaces

Linear Algebra Problem Sheet 5, Question 5.11 (local LA-questions.txt, pp. 25–29).

Let U, W be vector spaces over the same field K. Give $U \times W = \{(u, w) : u \in U, w \in W\}$ coordinatewise operations:

$$(u, w) + (u', w') = (u + u', w + w'), \qquad \lambda(u, w) = (\lambda u, \lambda w).$$

(a) Prove that $U \times W$ is a vector space over K. (b) Explain how this proves that K^n is a vector space over K.

> **Say:** An ordered pair is equal to another ordered pair precisely when both coordinates agree. That lets every axiom pass from the two factors to their product.

> 不要一开始把证明简写成“显然”；给加法结合与标量结合两个样例，再逐项标明零元、逆元、分配、单位。

(a) Addition and scalar multiplication are closed because they are closed in each factor. For $p = (u, w), q = (u', w'), r = (u'', w'')$, associativity and commutativity follow coordinatewise, for example

$$(p + q) + r = ((u + u') + u'', (w + w') + w'') = (u + (u' + u''), w + (w' + w'')) = p + (q + r).$$

The zero vector is $(0_U, 0_W)$, and the additive inverse of (u, w) is $(-u, -w)$. For $\lambda, \mu \in K$, the scalar axioms follow from those in U, W:

$$\lambda(p + q) = \lambda p + \lambda q, \qquad (\lambda + \mu)p = \lambda p + \mu p,$$

$$\lambda(\mu p) = (\lambda\mu)p, \qquad 1_K p = p.$$

Each equality is an equality of the two coordinates; for instance $\lambda(\mu p) = (\lambda(\mu u), \lambda(\mu w)) = ((\lambda\mu)u, (\lambda\mu)w)$. Thus all vector space axioms hold.

(b) A field K is a vector space over itself. Inductively, if K^m is a vector space, then so is $K^m \times K$, identified with K^{m+1} under the coordinatewise operations. Therefore K^n is a vector space for every positive integer n.

RESERVE / LOOKUP

LA5.12 · Complements need not be unique

Linear Algebra Problem Sheet 5, Question 5.12 (local LA-questions.txt, pp. 25–29).

Let $V = \mathbb{R}^2$, $U = \operatorname{span}\{e_1\}$. Is there a unique subspace W complementary to U in V, meaning $V = U \oplus W$?

> **Say:** A complement supplies the missing direction, but it does not have to be perpendicular. Check both spanning the whole space and having zero intersection.

> 不要把complement理解成唯一正交补；题目只要求直和补，给两条不同直线便足够反例。

No. Take $W_1 = \operatorname{span}\{e_2\}$ and $W_2 = \operatorname{span}\{e_1 + e_2\}$. They are distinct lines, and

$$(x, y)^T = xe_1 + ye_2 = (x - y)e_1 + y(e_1 + e_2).$$

Thus $U + W_1 = U + W_2 = \mathbb{R}^2$. Also $ae_1 = be_2$ implies $a = b = 0$, and $ae_1 = b(e_1 + e_2)$ implies $b = 0$, then $a = 0$. Hence $U \cap W_j = \{0\}$ for both j, proving that both are complements. More generally every line spanned by $(a, b)^T$ with $b \neq 0$ is a complement of U.

Coverage and source notes

The 55-minute route covers method families; the reserve bank provides the remaining problem-sheet answers for teacher lookup. It is not a claim that all subparts can be proved aloud in 55 minutes.

Item	Source	Use
P1	Pure PS5 A9(a–c), A10(a–c)	Main
P2	Pure PS5 B3(a–c)	Main
P3	Pure PS5 A2(a–b)	Main
P4	Pure PS5 A5(a–c), B6, B7	Main
P5	Pure PS5 A11	Main
P6	Pure PS5 A6(a–d)	Main
P7	Pure PS5 A1	Reserve / hints only where stated
P8	Pure PS5 A3	Reserve / hints only where stated
P9	Pure PS5 A4	Reserve / hints only where stated
P10	Pure PS5 A7(a–d); inverse functions / reserve	Reserve / hints only where stated
P11	Pure PS5 A8(a–c); inverse functions / reserve	Reserve / hints only where stated
P12	Pure PS5 A12(a–b); inverse functions / reserve	Reserve / hints only where stated
P13	Pure PS5 B1(a–c)	Reserve / hints only where stated
P14	Pure PS5 B2(a–c); inverse functions / reserve	Reserve / hints only where stated
P15	Pure PS5 B4(a–c)	Reserve / hints only where stated
P16	Pure PS5 B5	Reserve / hints only where stated
P17	Pure PS5 B8(a–b); due Thu 29 Oct 2026, 13:00; peer marking due Thu 5 Nov, 13:00	Reserve / hints only where stated

P18	Pure PS5 B9; inverse functions / reserve	Reserve / hints only where stated
P19	Pure PS5 B10(a–b); inverse functions / reserve	Reserve / hints only where stated
P20	Pure PS5 B11(a–d); inverse functions / reserve	Reserve / hints only where stated
P21	Pure PS5 B12(a–d); inverse functions / reserve	Reserve / hints only where stated
P22	Pure PS5 B13; inverse functions / reserve	Reserve / hints only where stated
P23	Pure PS5 B14; later material / reserve	Reserve / hints only where stated
P24	Pure PS5 C1, C2; reserve	Reserve / hints only where stated
P25	Pure PS5 C3(a–b); reserve	Reserve / hints only where stated
P26	Pure PS5 C4 (all four requests; original lettering duplicated); inverse functions / reserve	Reserve / hints only where stated
LA1	Linear Algebra tutorial guide, Week 5, pp. 12–13; examples supplied for the guide's discussion prompts.	Main
LA2	Linear Algebra tutorial guide, Week 5, p. 12, examples (i)–(iv).	Main
LA3	Linear Algebra tutorial guide, Week 5, p. 12, first basis example; real parameter domain made explicit.	Main
LA4	Linear Algebra tutorial guide, Week 5, p. 12, second basis example.	Main
LA5	Linear Algebra tutorial guide, Week 5, p. 12, third basis example; renamed V3 because the source repeats V2.	Main
LA6	Linear Algebra tutorial guide, Week 5, p. 12, warm-up.	Main
LA7	Linear Algebra tutorial guide, Week 5, p. 13; the source's condition $\|S\|<7$ is preserved, then strengthened explicitly.	Main
LA8	Linear Algebra tutorial guide, Week 5, p. 13, basis reduction/extension discussion; concrete examples supplied.	Main
LA9	Linear Algebra tutorial guide, Week 5, p. 13, challenging question and footnote 7; missing y parameter in source footnote corrected.	Reserve / hints only where stated
LA10	Linear Algebra tutorial guide, Week 5, p. 13, End and hand-in boundary.	Reserve / hints only where stated

LA5.2	Linear Algebra Problem Sheet 5, Question 5.2 (local LA-questions.txt, pp. 25–29).	Reserve / hints only where stated
LA5.3	Linear Algebra Problem Sheet 5, Question 5.3 (local LA-questions.txt, pp. 25–29).	Reserve / hints only where stated
LA5.4	Linear Algebra Problem Sheet 5, Question 5.4 (local LA-questions.txt, pp. 25–29).	Reserve / hints only where stated
LA5.5	Linear Algebra Problem Sheet 5, Question 5.5 (local LA-questions.txt, pp. 25–29).	Reserve / hints only where stated
LA5.6	Linear Algebra Problem Sheet 5, Question 5.6 (local LA-questions.txt, pp. 25–29).	Reserve / hints only where stated
LA5.7	Linear Algebra Problem Sheet 5, Question 5.7 (local LA-questions.txt, pp. 25–29).	Reserve / hints only where stated
LA5.8	Linear Algebra Problem Sheet 5, Question 5.8 (local LA-questions.txt, pp. 25–29).	Reserve / hints only where stated
LA5.9	Linear Algebra Problem Sheet 5, Question 5.9 (local LA-questions.txt, pp. 25–29).	Reserve / hints only where stated
LA5.10	Linear Algebra Problem Sheet 5, Question 5.10 (local LA-questions.txt, pp. 25–29).	Reserve / hints only where stated
LA5.11	Linear Algebra Problem Sheet 5, Question 5.11 (local LA-questions.txt, pp. 25–29).	Reserve / hints only where stated
LA5.12	Linear Algebra Problem Sheet 5, Question 5.12 (local LA-questions.txt, pp. 25–29).	Reserve / hints only where stated

Pure source: Florian Bouyer, “Intro to Pure - Week 5”, 16 October 2026, with Problem Sheet 5 and its solution attachment, retrieved from the university mailbox on 22 October. Additional Linear reserve questions 5.2–5.12 use the previously downloaded 2026 problem-sheet compilation; its stale hand-in date is not used. Linear source: Rachael Carey, “Re: Linear algebra marking”, 20 October 2026, with Linear Algebra tutorial ideas 2026, Week 5, pp. 12–13. Statements have been rechecked independently; obvious source typographical errors are corrected where identified.

Pure PS5 B8 is due 29 October at 13:00 for peer marking; the bank supplies only starting hints. The Linear Week 5 hand-in is due in Week 7 and is also hints only. Do not distribute the teacher answer bank to students. The companion student pack contains main questions and a small reserve selection without solutions.

> 未重新核对 Blackboard 当日的所有改动，也未确认周四线代实际讲完的位置；本版以已读教师邮件和附件为范围依据。视频内容不作为数学题目来源。